

PING locks E rate $\approx 10\times$ below COBA

exp025 · 30 May 2026

The trained networks this entry uses are produced once in the shared training hub, exp022 (Training), and reused here rather than retrained.

Abstract

Head-to-head comparison of COBA (recurrent inhibitory loop disabled) against PING (loop active) on MNIST under matched architecture and training recipe. PING locks the hidden E rate to ≈ 17 Hz while COBA runs at ≈ 114 Hz; accuracy is within a few points across the two, so the loop buys better than an order-of-magnitude per-spike economy. The rate-vs-accuracy frontier traced by sweeping the hidden-E rate ceiling regulariser shows the floor is structural, not a trade-off the optimiser can navigate away.

Method

Training recipe (canonical / medium tier):

Parameter	Value
Integration timestep Δt	0.1 ms
Trial duration T	200 ms
MNIST training pool (7,000 official-training samples)	6,300 optimizer-training / 700 validation; downstream inference uses a fixed 1,000-image subset of the official test partition
Epochs	50

Checkpoint-backed rates, rasters, and input-scaling interventions use the final-epoch checkpoint because this experiment characterizes the trained network’s endpoint dynamics, rather than deployment performance of the validation-selected classifier.

Two configurations of the same COBANet architecture, differing only in whether the $E \rightarrow I \rightarrow E$ inhibitory loop is active. **COBA** (*-ei-strength 0*) disables the loop; excitatory cells drive each other but receive no structured inhibition. **PING** (*-ei-strength 1*) enables the loop, producing pyramidal-interneuron gamma (PING) oscillations at a cadence set by τ_{AMPA} and τ_{GABA} .

Architecture. $N_E = 1024$ excitatory cells, $N_I = 256$ inhibitory cells, single hidden layer. Input: 784 channels (MNIST pixels), Poisson-encoded at 25 Hz peak rate. Readout: mem-mean (time-averaged E spike vector projected through a trained linear layer W_{out} ; see ar006 for the full readout specification). Dale’s law enforced.

What is trainable. Only the input weights W_{in} (784×1024 , 95% sparse) and the readout W_{out} (1024×10). The recurrent weights W_{ee} (fixed at zero, since Börgers-style PING needs no E→E coupling), W_{ei} , and W_{ie} are initialised once and held *requires_grad = False*. The synaptic time constants τ_{AMPA} , τ_{GABA} are module-level constants. 813k trainable parameters out of 2.4M total.

Training. Adam, $\text{lr} = 4 \times 10^{-4}$, batch size 256, gradient norm clipped to 1.0. Cross-entropy loss on 10-class MNIST (definition in ar006). The voltage-gradient damping is 1 for COBA and 1000 for PING, where recurrent backpropagation requires stabilisation. Every frontier condition is trained with three independent seeds (42, 43, 44); plotted points are across-seed means with standard errors.

Recipe difference. COBA and PING share $W_{\text{in}} \sim \mathcal{N}(0.9, 0.09)$ at 95% sparsity and the same directly specified readout initializer. COBA disables the recurrent loop; PING enables it and uses the loop-specific gradient damping.

Activity regulariser. To probe the rate axis, the training loss adds a soft upper bound on each presentation’s population-mean hidden-E firing rate:

$$r_b = \frac{1}{N_E T} \sum_{n \in E} z_{bn}, \quad L_{\text{rate}} = \frac{\lambda}{B} \sum_b \text{ReLU}(r_b - r_{\text{max}})^2,$$

with r_b the population-mean hidden-E firing rate for presentation b and r_{max} its ceiling in hertz. Presentations at or below the ceiling contribute zero. The normalized penalty uses $\lambda = 0.041$. We sweep off, 25, 10, 5, 2.5, and 1 Hz, giving twelve (model, r_{max}) conditions and 36 trained cells across seeds.

Rate-floor decomposition. The affine law $r_E = p \cdot f_\gamma$ (exp041, exp046) factors the E rate into per-cycle participation p and gamma frequency f_γ . Both are measured at one explicitly representative final-epoch checkpoint (seed 42) per condition: f_γ from the Welch PSD peak of the E-population trace, p via I-burst peak detection and per-(cell, cycle) spike counting (style of exp046).

Basin and landscape probes. To test basin attractivity, four input-coupling conditions are each trained with seeds 42, 43, and 44 (12 PING networks total), using parent means of 0.05, 0.1, 0.3, and 0.9 for the expected summed W_{in} initialization before lower clamping and fan-in normalization ($r_{\text{max}} = 1$ Hz from epoch 0; Figure 3). Exp022 owns these TR-07 trainings; this experiment aggregates their checkpoints. To map the loss landscape around the operating point, each network trained at the tightest ceiling ($r_{\text{max}} = 1$ Hz) has its W_{in} scaled by a common scalar $s \in [0.05, 3]$ at inference with all other weights frozen, metrics averaged over the same fixed 1,000-image held-out subset at 24 values of s (Figures 4–5).

Results

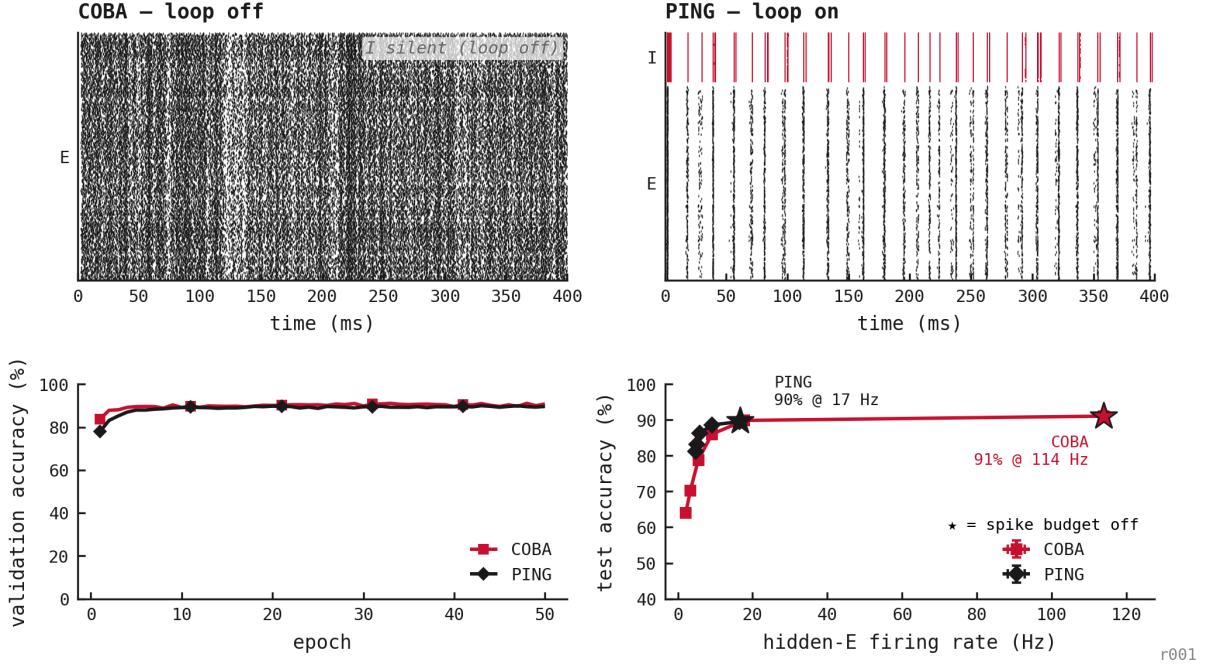


Figure 1: The headline comparison in one 2×2 frame (rasters, learning, and the accuracy–rate frontier together). **Top:** trained-baseline single-trial rasters on the same MNIST digit 0 input. COBA ($-ei\text{-strength } 0$) fires densely and asynchronously at ≈ 114 Hz (I silent, loop off); PING ($-ei\text{-strength } 1$) fires in gamma bands at ≈ 17 ms cadence (≈ 17 Hz per E cell) with synchronous I bursts (red) above E (black), on the same architecture, parameter count, and recipe, with only PING’s recurrent $E \leftrightarrow I$ matrices non-zero. **Bottom left:** validation accuracy per epoch (mean over three seeds); both reach $\approx 90\%$, so both learn the task. **Bottom right:** the accuracy–rate frontier across hidden-E rate ceilings (mean $\pm$ SEM across three seeds at every point). PING sits up-and-left of COBA, the same accuracy at a fraction of the hidden-E rate, with the unpenalised operating points starred and labelled (PING 90% at ≈ 17 Hz, COBA 91% at ≈ 114 Hz). COBA red, PING black; E black and I red in the rasters. *The headline of this entry: gamma gating buys an order-of-magnitude per-spike economy at matched accuracy.*

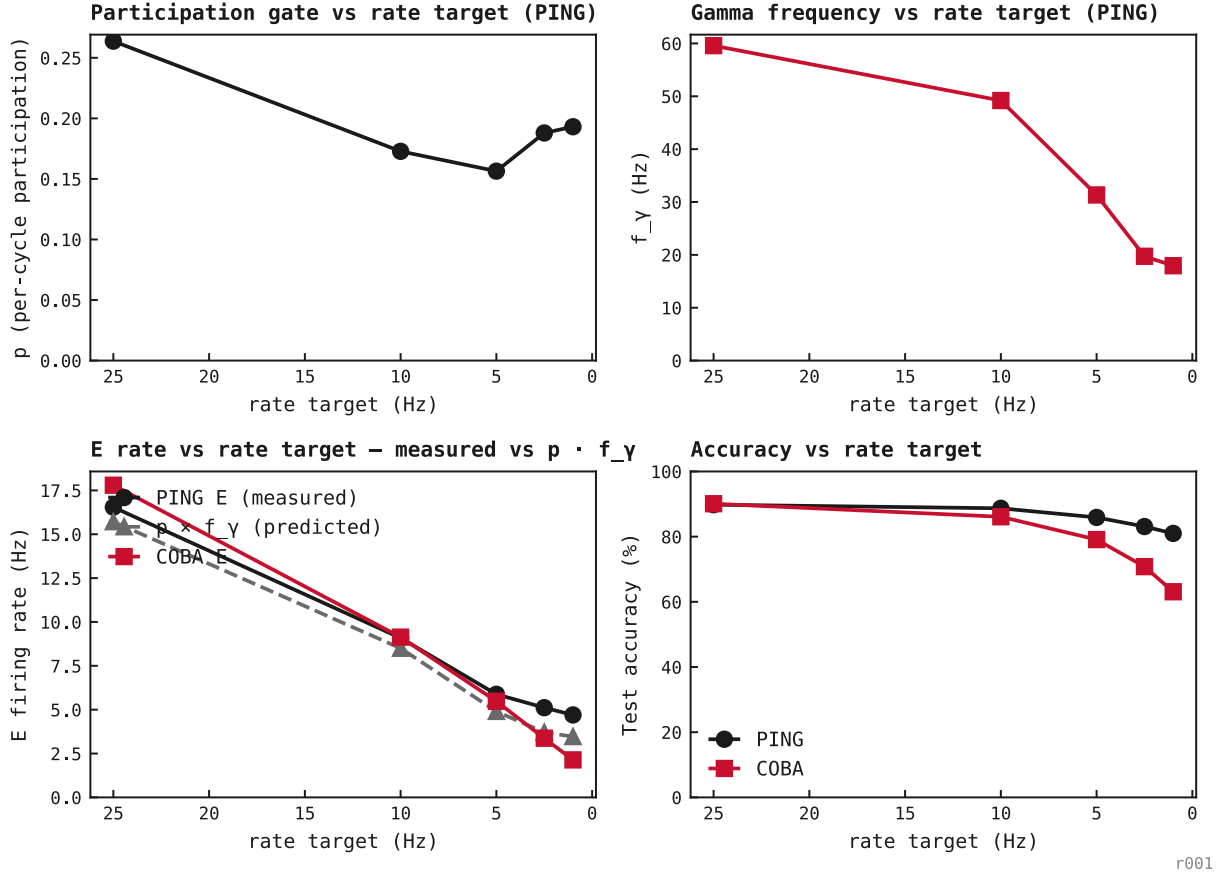


Figure 2: Five representative seed-42 PING rate-target checkpoints, 256 test trials each. p stays in 0.16–0.26 across the entire sweep (the architecture protects the participation gate), while f_γ slides from ≈ 60 Hz to ≈ 18 Hz as the penalty tightens. The grey dashed $p \cdot f_\gamma$ curve overlays the measured E rate within 27%; the rate change is entirely in f_γ . PING accuracy holds at 81–90% the whole way; COBA’s collapses from 90% to 63%.

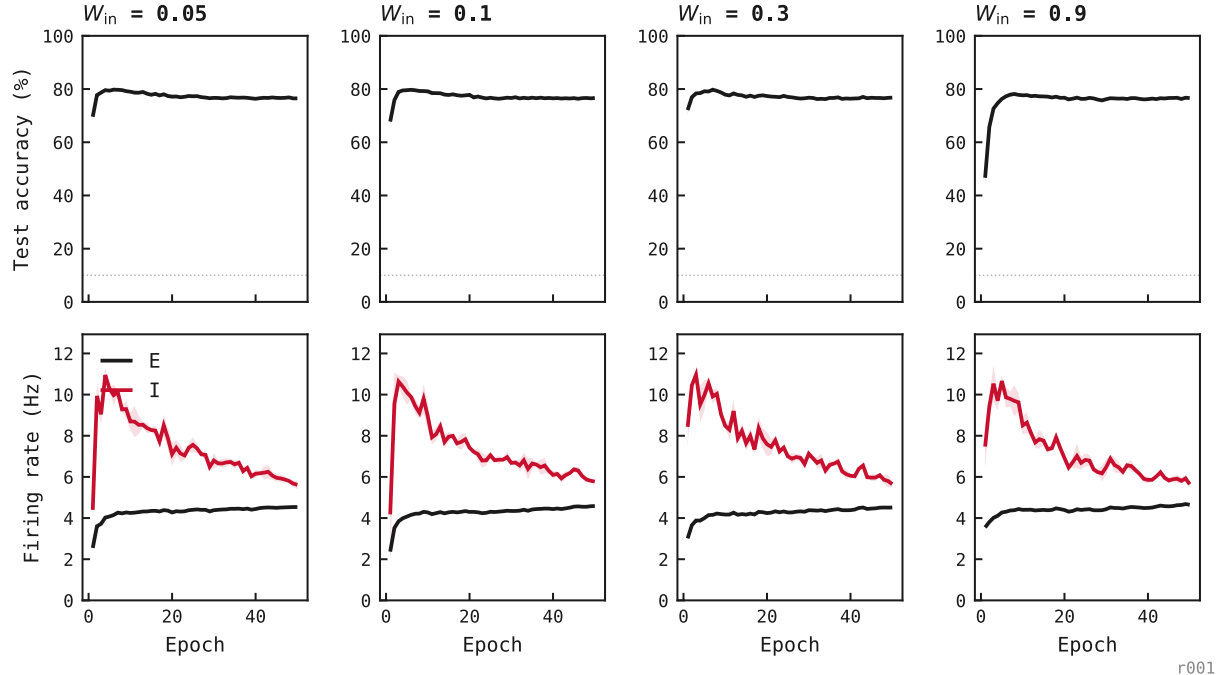
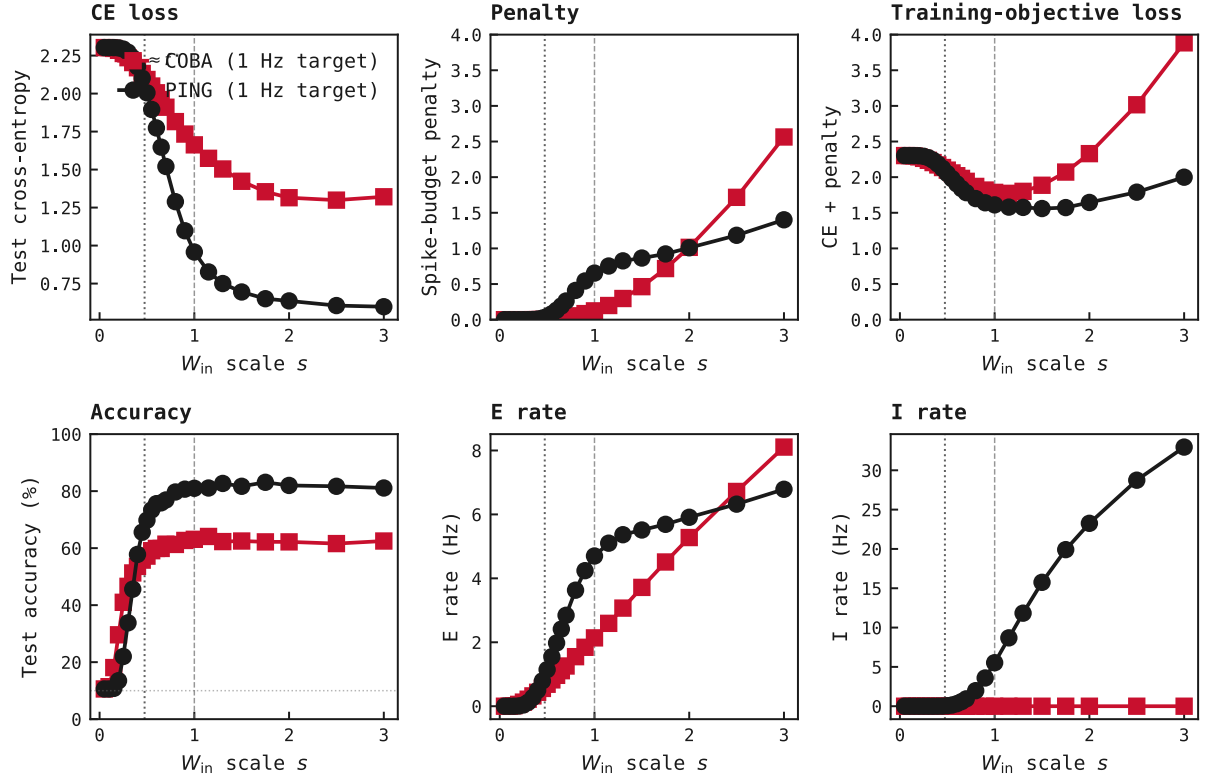


Figure 3: Across-seed mean per-epoch training traces for four input summed-coupling parent means (seeds 42–44, $r_{\max} = 1$ Hz from epoch 0), one condition per column; shaded bands are SEM. Top: test accuracy. Bottom: test-set E (black) and I (red) firing rates. The comparison tests whether lower initial input drive delays or prevents recruitment of the inhibitory loop. Final mean accuracies: 76.5% / 76.6% / 76.8% / 76.6%; final I rates: 5.6 / 5.8 / 5.6 / 5.6 Hz.



r001

Figure 4: Inference-time W_{in} scale sweep on the two networks trained under the tightest ceiling ($r_{max} = 1$ Hz); every W_{in} weight multiplied by a common scalar s , all other weights frozen, 24 values of $s \in [0.05, 3]$. Top row: CE loss, activity penalty L_{rate} , total objective CE + L_{rate} . Bottom row: test accuracy (chance dotted), E rate, I rate. PING black, COBA red. Vertical dashed line at $s = 1$ marks the trained operating point; dotted line marks $\approx f^*$, PING’s recruitment cliff. Loss panels clipped at 4; COBA’s penalty reaches ≈ 3 at $s = 3$ (rate² scaling).

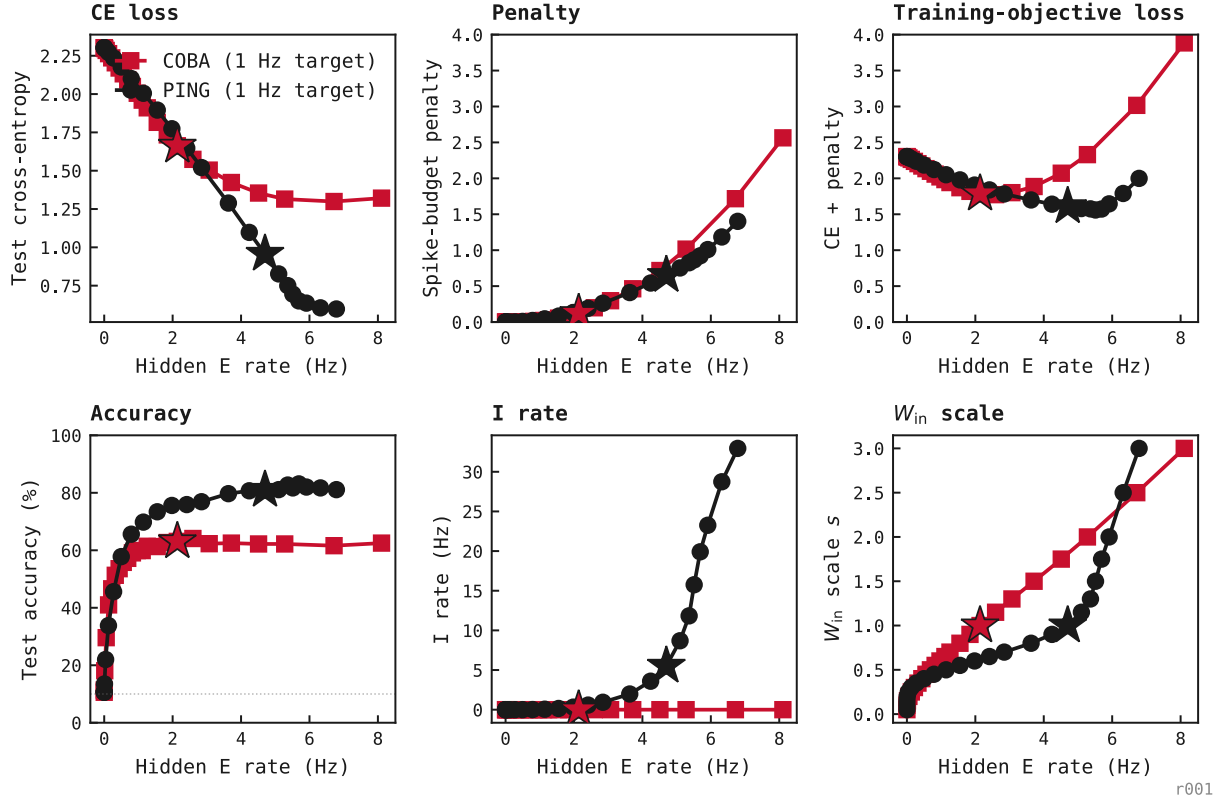


Figure 5: The same 24-point sweep as Figure 4, re-projected with hidden E rate on the x-axis. Filled stars mark each cell’s trained operating point ($s = 1$): PING at $E \approx 4.7$ Hz, COBA at $E \approx 2.1$ Hz. PING’s accuracy reaches its plateau ($\approx 83\%$) just past its trained operating point; COBA climbs slowly and only reaches $\approx 63\%$ even at $E \approx 8$ Hz.